جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

The Severity and Impact of Double-Parking Incidents on Cairo Drivers

Submitted by:

Eyad Mohamed Fakeih	ID: 24-101090
Kareem Ali Nabil	ID: 24-101106
Abdurrahman Ehab	ID: 24-101109
Ali Ahmed ElSayed	ID: 24-101429

Introduction

Cairo is one of the most densely populated cities in the world, with over 4.2 million registered vehicles navigating its streets daily. Among the many traffic challenges faced by Cairo drivers, double-parking stands out as a persistent and uniquely disruptive problem. Unlike general traffic congestion, double-parking creates a direct and immediate blockage for individual drivers, trapping them completely until the blocking car owner returns or is located.

This problem is not merely a matter of inconvenience. It represents a measurable loss of time, a documented source of psychological stress, and a trigger for social conflict. Drivers who find themselves blocked have limited options: they may honk, ask bystanders for help, call the traffic police, or simply wait. None of these solutions are efficient or reliable.

The motivation for this study stems from a real entrepreneurial initiative called Efta7 ElTareeq, which proposed an anonymous digital notification system to resolve double-parking incidents in under 60 seconds. Before any such solution can be built and validated, the actual severity of the problem must first be measured rigorously. This report does exactly that.

By collecting survey data from 54 Cairo drivers across multiple age groups, genders, and city areas, this study aims to quantify the frequency, duration, and emotional impact of double-parking incidents and determine whether these impacts vary across demographic and geographic groups.

Research Question

How severely does double-parking affect Cairo drivers in terms of time lost, stress experienced, and frequency of incidents — and does this impact vary by age, gender, or driving area?

Hypothesis

This study tests the following hypothesis with three specific testable predictions:

Cairo drivers experience double-parking incidents at least 4 times per month, with an average wait time exceeding 10 minutes per incident, and female drivers report significantly higher stress levels than male drivers during these incidents.

Prediction 1: The mean number of double-parking incidents experienced per month is greater than or equal to 4.

Prediction 2: The mean wait time per blocking incident exceeds 10 minutes.

Prediction 3: Female drivers report a higher average stress level than male drivers.

Methodology

Population of Interest

Urban car drivers in Greater Cairo (Cairo and Giza governorates), aged 18 and above, who drive at least once per week. This population was estimated at approximately 4.2 million vehicle owners based on published traffic authority data.

Sampling Method

Convenience sampling was used by distributing the survey digitally through WhatsApp groups, university networks, and social media platforms. This method was chosen because there is no publicly accessible registry of Cairo drivers that would allow for random sampling. Convenience sampling is the most practical and ethical approach for an academic survey of this scale.

To improve the representativeness of the sample, the survey was deliberately shared across mixed-age WhatsApp groups, family networks, and professional communities — not just university students — in order to capture responses from a diverse range of ages, genders, and driving areas.

A total of 54 valid responses were collected over one week, surpassing the minimum threshold of 40 responses required for meaningful statistical analysis.

Survey Questions

- **Q1.** What is your gender? (Male / Female / Prefer not to say)
- **Q2.** What is your age group? (18-25 / 26-35 / 36-45 / 46 and above)
- **Q3.** How many years have you been driving in Cairo? (Open number)
- **Q4.** How many days per week do you typically drive in Cairo? (1-2 / 3-4 / 5-6 / Every day)
- **Q5.** On average, how many times per month are you blocked by a double-parked car? (Open number)
- **Q6.** When you are blocked, how many minutes do you typically wait before the situation is resolved? (Open number)
- **Q7.** How many times in the past month were you late to a destination because of double parking? (Open number)
- **Q8.** On a scale from 1 to 10, how stressful do you find double-parking incidents?
- **Q9.** What do you usually do when you are blocked by a double-parked car? (Multiple choice)
- **Q10.** In which area of Cairo do you drive most frequently? (Multiple choice)

Bias Identification and Mitigation

Bias Type	How It Appears	How We Mitigated It
Survivorship Bias	People who never get blocked may ignore the survey	Q5 includes 0 as a valid answer so all drivers can respond
Social Desirability Bias	People may exaggerate stress to appear relatable	Survey is fully anonymous with no names or emails collected
Sampling Bias	Mostly university networks may over-represent young respondents	Survey shared in mixed-age WhatsApp groups
Recall Bias	Respondents may not accurately remember exact wait times	All time and frequency questions ask for averages, not specific incidents

Analysis

To analyze the data collected from this survey, we calculated descriptive statistics including the mean, median, and mode for all numerical variables. We also created visual representations of the data using a variety of chart types to identify trends, patterns, and relationships across demographic and geographic groups.

Descriptive Statistics Summary

Variable	Mean	Median	Mode
Years Driving	4.43	3.0	2
Incidents / Month	4.56	3.0	3
Wait Time (min)	14.5	15.0	15
Times Late / Month	3.2	2.0	0
Stress Level (1-10)	7.26	8.0	10

The descriptive statistics reveal several important patterns. The mean number of incidents per month is 4.56, confirming the first prediction of our hypothesis. The mean wait time is 14.5 minutes, well above the 10-minute threshold predicted. The stress level mean of 7.26 out of 10 — with a mode of 10 — indicates that double-parking is perceived as an extremely stressful experience by the majority of Cairo drivers.

Chart 1: Distribution of Stress Levels

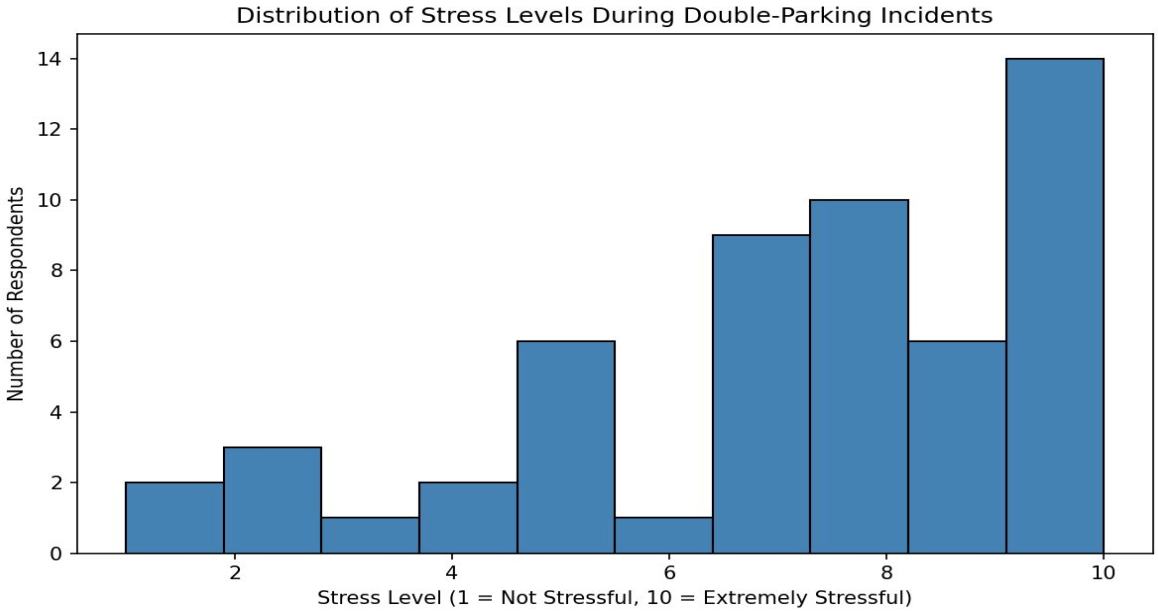

Figure 1: Distribution of Stress Levels During Double-Parking Incidents

The histogram shows a clear right-skewed distribution of stress levels, with the majority of respondents reporting stress levels between 7 and 10. The tallest bar falls at stress level 10, with 14 respondents — more than a quarter of the entire sample — selecting the maximum possible stress rating. This distribution confirms that double-parking is not merely a minor inconvenience for Cairo drivers but a genuinely distressing experience.

Notably, very few respondents reported low stress levels, with only 6 respondents selecting a score of 4 or below. This suggests that regardless of how frequently a driver encounters double-parking, the experience itself is consistently perceived as highly stressful across the population.

Chart 2: Driver Behavior When Blocked

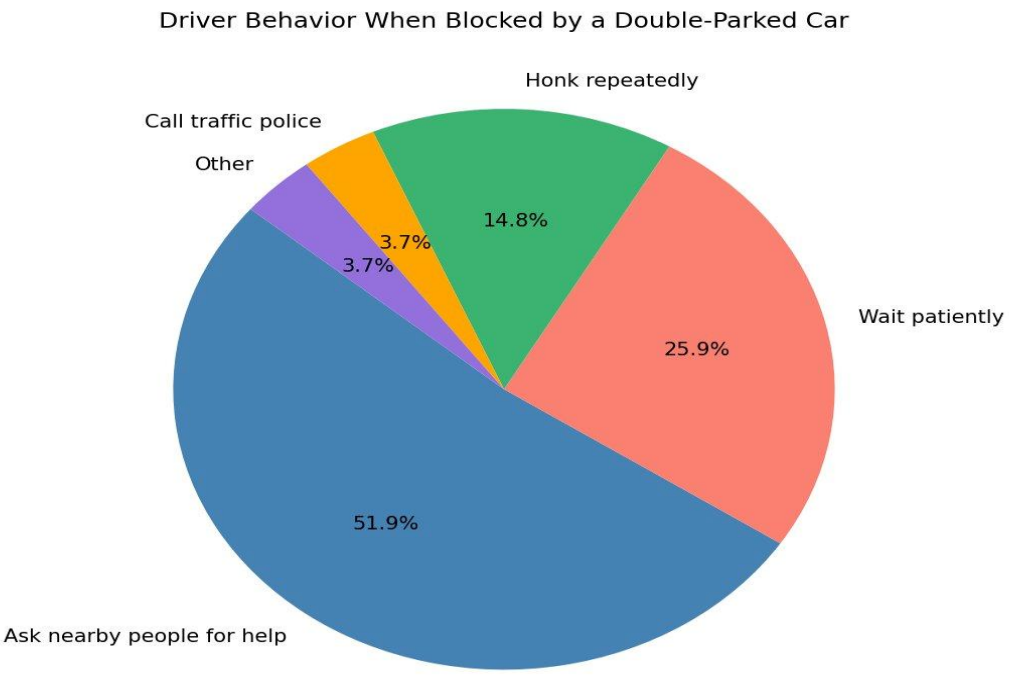

Figure 2: Driver Behavior When Blocked by a Double-Parked Car

The pie chart reveals that the dominant response among Cairo drivers when blocked is to ask nearby people for help, accounting for 51.9% of all respondents. This is followed by waiting patiently at 25.9%, and honking repeatedly at 14.8%. Notably, only 3.7% of respondents call the traffic police, suggesting that formal intervention channels are largely ineffective or inaccessible in practice.

The overwhelming preference for informal community-based solutions — asking bystanders to locate the blocking car owner — highlights a critical gap in the current system. Cairo drivers have collectively developed informal workarounds because no standardized, efficient, or anonymous communication channel exists between blocked and blocking drivers.

Chart 3: Average Stress Level by Gender

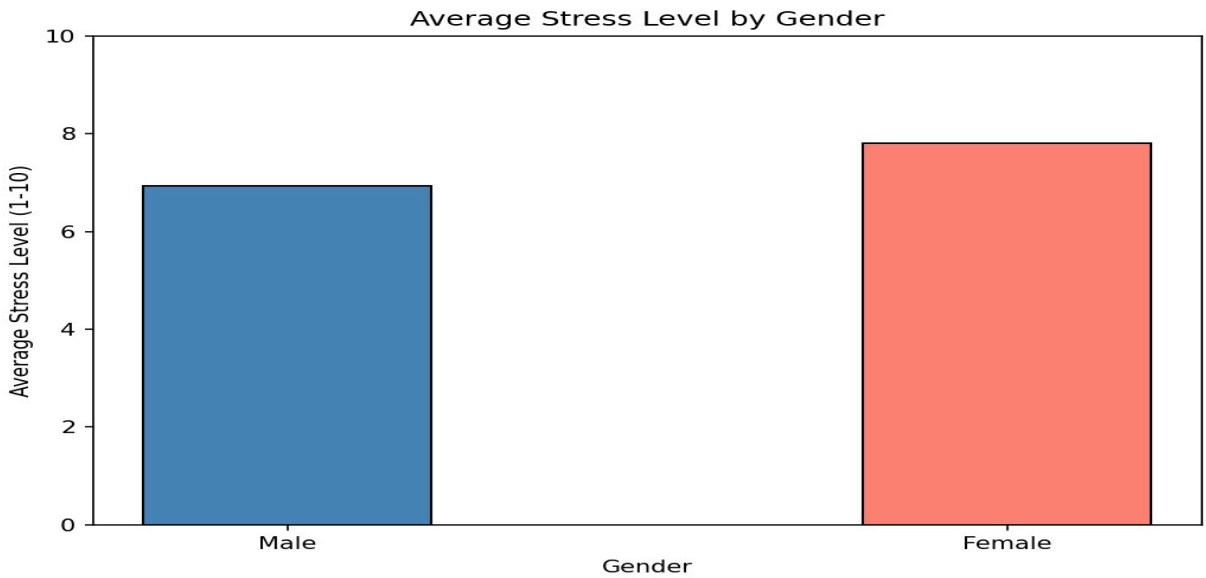

Figure 3: Average Stress Level by Gender

The bar chart directly tests the third prediction of our hypothesis. Female drivers reported an average stress level of 7.80, compared to 6.94 for male drivers — a difference of 0.86 points on the 10-point scale. This confirms that female drivers experience measurably higher stress during double-parking incidents than their male counterparts.

This finding is consistent with broader urban mobility research, which has documented that female drivers in high-density cities often face compounded stressors including safety concerns, harassment risks, and the additional pressure of managing time-sensitive responsibilities. The gap in stress levels between genders reinforces the need for anonymous and privacy-preserving solutions to the double-parking problem.

Chart 4: Average Wait Time by Age Group

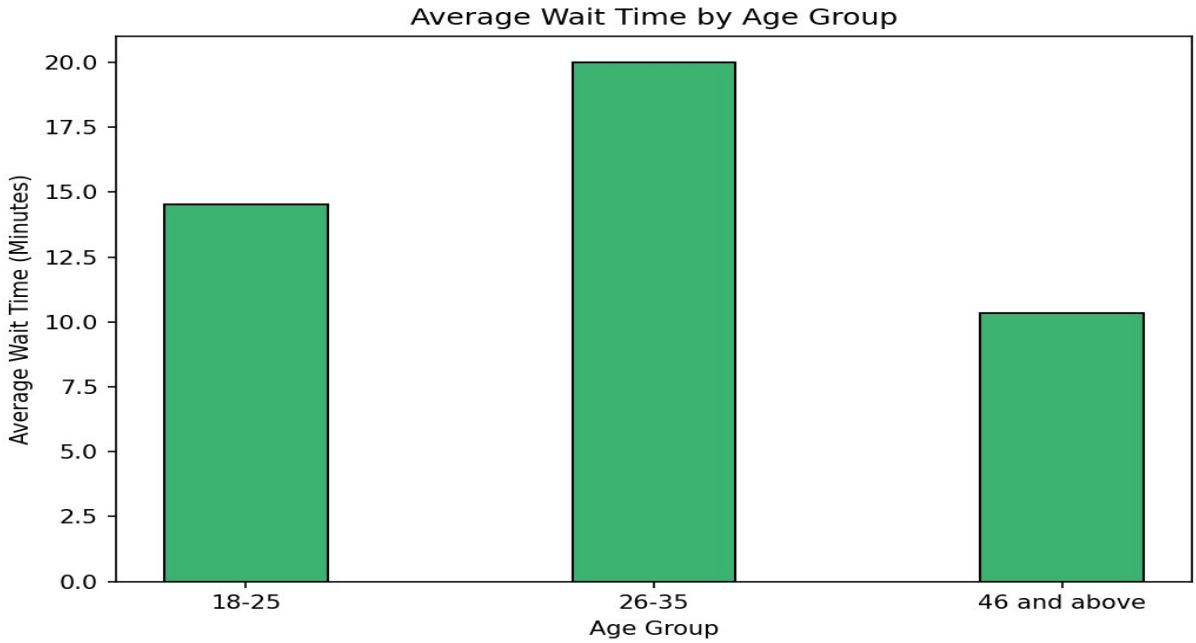

Figure 4: Average Wait Time by Age Group

The bar chart reveals a notable pattern across age groups. Drivers aged 26-35 experience the longest average wait time at 20 minutes, significantly higher than the 18-25 group at 14.53 minutes and the 46 and above group at 10.33 minutes. This suggests that middle-aged working professionals — who are most likely to drive daily through high-density commercial and residential areas — bear the greatest time cost from double-parking incidents.

The 46 and above age group reports the shortest wait time, which may reflect greater experience navigating urban traffic situations, more flexible schedules, or a tendency to drive in less congested areas. These findings suggest that targeted solutions should prioritize the needs of the 26-35 demographic, who suffer the most significant time impact.

Chart 5: Average Incidents Per Month by Area

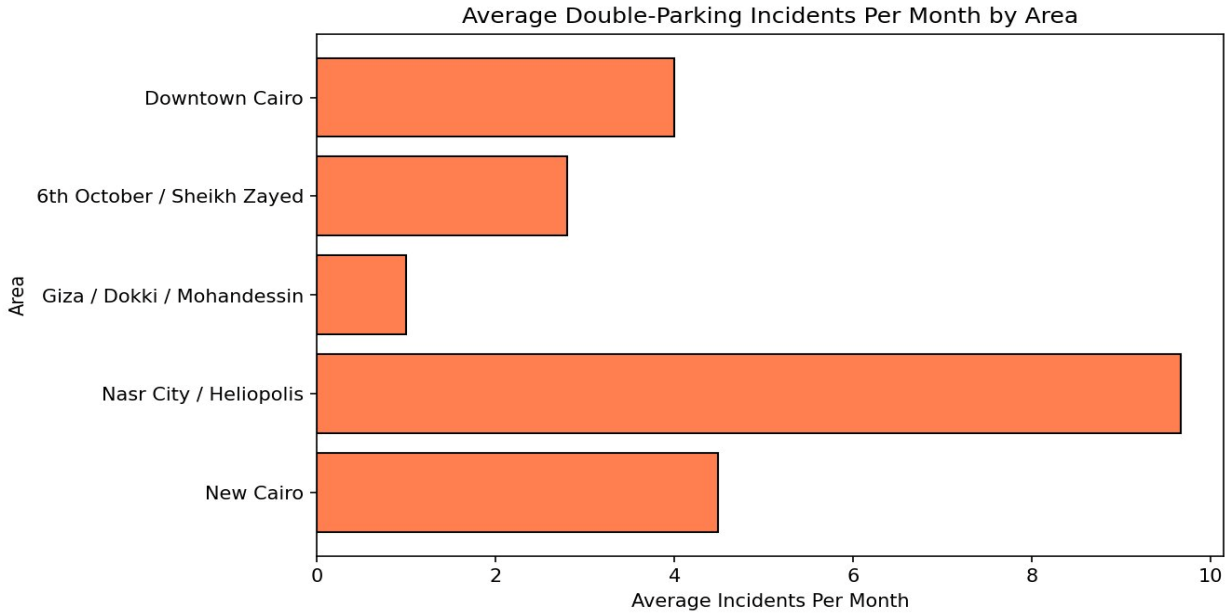

Figure 5: Average Double-Parking Incidents Per Month by Area

The horizontal bar chart clearly identifies Nasr City and Heliopolis as the most severely affected area in Cairo, with an average of nearly 10 double-parking incidents per month — more than double the rate observed in New Cairo and Downtown Cairo. This is consistent with the high population density, commercial activity, and narrow street layouts characteristic of these older urban districts.

In contrast, areas such as Giza, Dokki, and Mohandessin report a surprisingly low average of approximately 1 incident per month. It is important to note that sample sizes vary significantly across areas, with 35 of 54 respondents driving primarily in New Cairo. Future research with more geographically balanced sampling would provide more reliable area-level comparisons.

Chart 6: Incidents Per Month vs Stress Level (Scatter Plot)

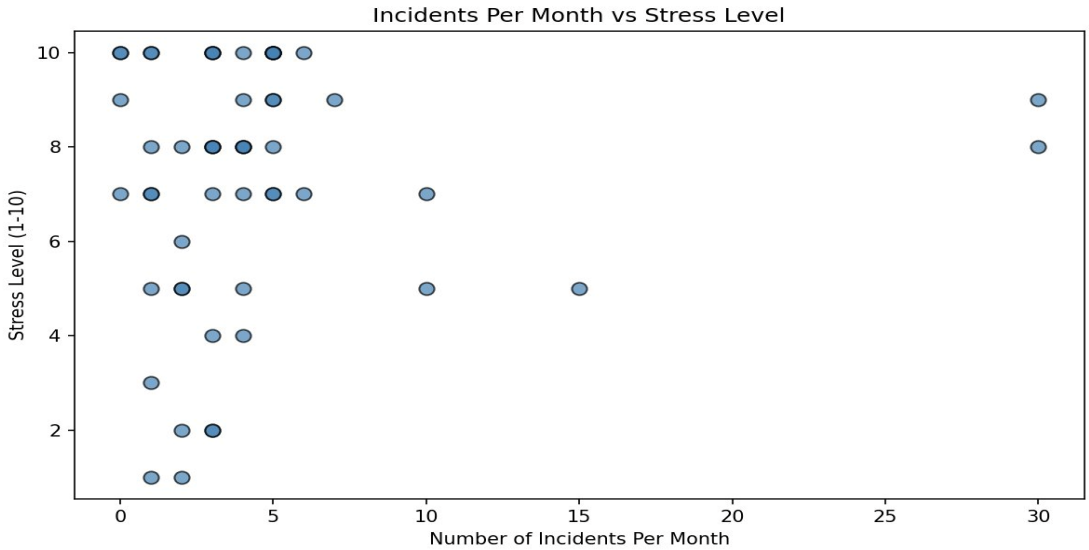

Figure 6: Scatter Plot of Incidents Per Month vs Stress Level

The scatter plot reveals an important and counterintuitive finding: high stress levels are reported across all ranges of incident frequency. Drivers who report zero incidents per month still rate their stress level at 8, 9, or 10. This suggests that stress from double-parking in Cairo is not primarily driven by how often it occurs, but rather by the severity and helplessness of the experience each time it does occur.

The wide vertical spread of points at low incident counts further confirms that stress is highly variable and individually subjective. Even a single double-parking incident per month can be rated as maximally stressful, indicating that the emotional impact of being trapped with no clear resolution mechanism is the core driver of distress — not mere frequency.

Chart 7: Wait Time Distribution by Gender (Box Plot)

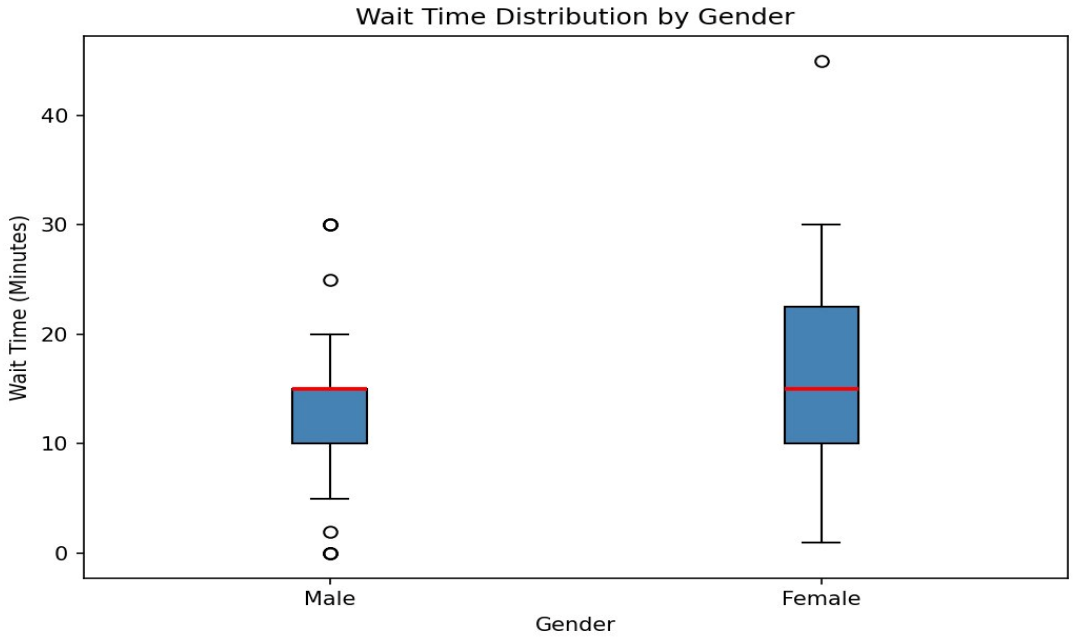

Figure 7: Box Plot of Wait Time Distribution by Gender

The box plot provides a more complete picture of wait time distribution beyond simple averages. Both male and female drivers share a similar median wait time of approximately 15 minutes, shown by the red line in each box. However, the female box is noticeably taller, indicating a wider spread — female respondents show greater variability in their wait times, with the upper quartile reaching approximately 23 minutes compared to 20 minutes for males.

The female box plot also shows a notable outlier at 45 minutes, representing an extreme case of a very long wait. Male respondents show multiple outliers at 25 and 30 minutes. These outliers are important because they represent real worst-case scenarios that drivers face, and they highlight that while median wait times are similar between genders, female drivers are more exposed to prolonged blocking incidents.

Conclusion

This study set out to measure the severity and impact of double-parking incidents on Cairo drivers across three testable predictions. Based on the data collected from 54 respondents, the findings are as follows:

Prediction 1 — Confirmed: The mean number of incidents per month was 4.56, exceeding the predicted threshold of 4 incidents. Cairo drivers are blocked by double-parked cars frequently, with the median being 3 incidents per month and the mode being 3.

Prediction 2 — Confirmed: The mean wait time was 14.5 minutes per incident, significantly above the 10-minute threshold. With a median of 15 minutes and a mode of 15 minutes, the data consistently shows that blocking incidents consume a substantial portion of drivers' time.

Prediction 3 — Confirmed: Female drivers reported a mean stress level of 7.80 compared to 6.94 for male drivers, confirming that female drivers experience measurably higher stress. All three predictions of the hypothesis are supported by the data.

Beyond the hypothesis, the analysis revealed several additional findings of note. Nasr City and Heliopolis emerged as the most severely affected area, with nearly 10 incidents per month on average. The 26-35 age group experiences the longest wait times, suggesting working professionals bear the greatest time cost. The dominant behavioral response — asking nearby people for help — demonstrates the absence of a standardized communication channel between drivers. Finally, the scatter plot revealed that stress is driven by the nature of the experience rather than its frequency alone, with even infrequent sufferers reporting maximum stress levels.

Taken together, these findings paint a clear picture: double-parking in Cairo is a frequent, time-consuming, and highly stressful problem that affects drivers across all demographics, with particular severity for female drivers and those in older, denser urban districts.

Critical Reflection

While this study produced meaningful and consistent findings, several limitations and potential issues were identified during the data collection and analysis process.

Geographic imbalance: 35 out of 54 respondents (64.8%) reported driving primarily in New Cairo and the Fifth Settlement. This geographic concentration means that area-level comparisons, particularly for Downtown Cairo and Giza, are based on very small sample sizes (3 and 5 respondents respectively). The strong result for Nasr City and Heliopolis, while notable, was also based on only 6 respondents. Future research should ensure more geographically balanced sampling to produce reliable area-level insights.

Age group imbalance: The vast majority of respondents fell in the 18-25 age group, which is expected given the convenience sampling method through university and social media networks. The 36-45 age group was entirely absent from the data, which limits the completeness of the age-based analysis. This is a direct consequence of the sampling method and should be acknowledged as a limitation.

Recall bias: Despite asking for averages rather than specific incidents, respondents may still have provided estimates influenced by recent memorable experiences rather than true long-term averages. A driver who was blocked for 45 minutes last week may overestimate their typical wait time as a result.

Self-reporting limitations: Stress levels and behavioral responses are self-reported and subjective. Two drivers with identical experiences may rate their stress very differently based on personality, tolerance, and context. The 1-10 stress scale, while useful for comparison, does not capture the full complexity of the emotional experience.

Hidden bias discovered post-collection: Upon reviewing the data, a concentration of respondents from one geographic cluster (New Cairo) was identified. This was not anticipated during the survey design phase and suggests that future studies should include quota-based sampling targets per area to ensure representativeness.

Despite these limitations, the study successfully demonstrates that double-parking is a statistically significant and emotionally impactful problem for Cairo drivers, and the findings provide a solid evidence base for further research and potential solution development.